ATHARVA UNIVERSITY
MUMBAI

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SMART TRANSFORMER HEALTH MONITORING SYSTEM

Design and Simulation using MATLAB / Simulink

Mini Project under Development by Atharva University

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[Designation]
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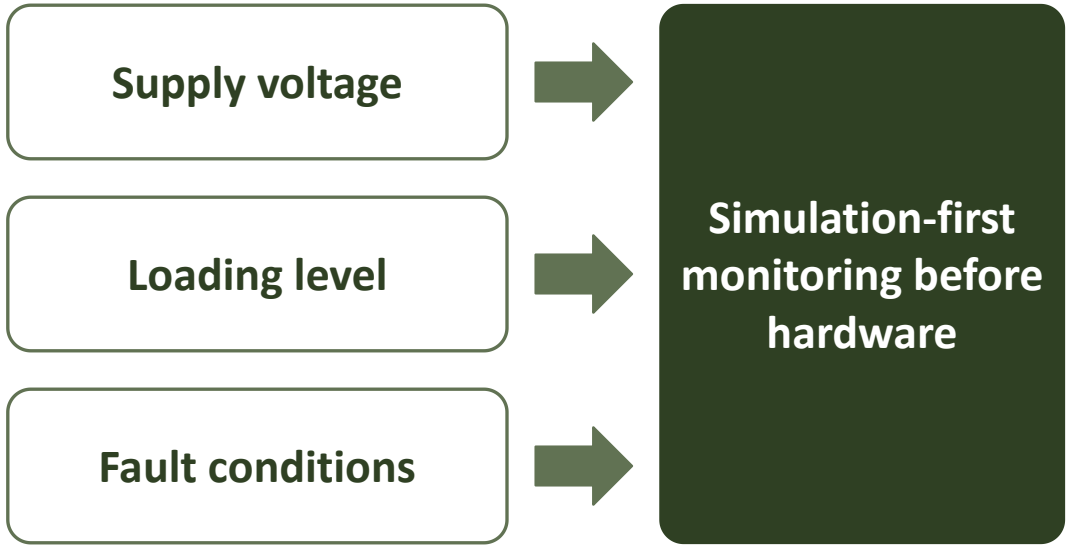
PROBLEM STATEMENT

Transformer operating conditions change with supply voltage, loading and faults. A monitoring system must detect these changes and present useful condition information.

Developing and testing the monitoring logic directly on hardware is costly and hard to troubleshoot.

Aim

Build a three-phase Simulink model that monitors electrical parameters, generates temperature and vibration indicators, flags abnormal conditions and shows results on a dashboard.



Repeatable test conditions let the team verify calculated parameters and the monitoring logic before any sensor board is built.

INTRODUCTION

A power transformer transfers energy by electromagnetic induction and is central to generation, transmission and distribution. Condition monitoring spots abnormal operation through continuous measurement and diagnostic logic.

5 kVA	400 / 230 V	Δ - grounded Y
Rating	Line-to-line ratio	Winding connection

Scope: Simulink and Simscape Electrical simulation only. Physical sensor board, ESP32 firmware and IoT link are out of scope this semester.

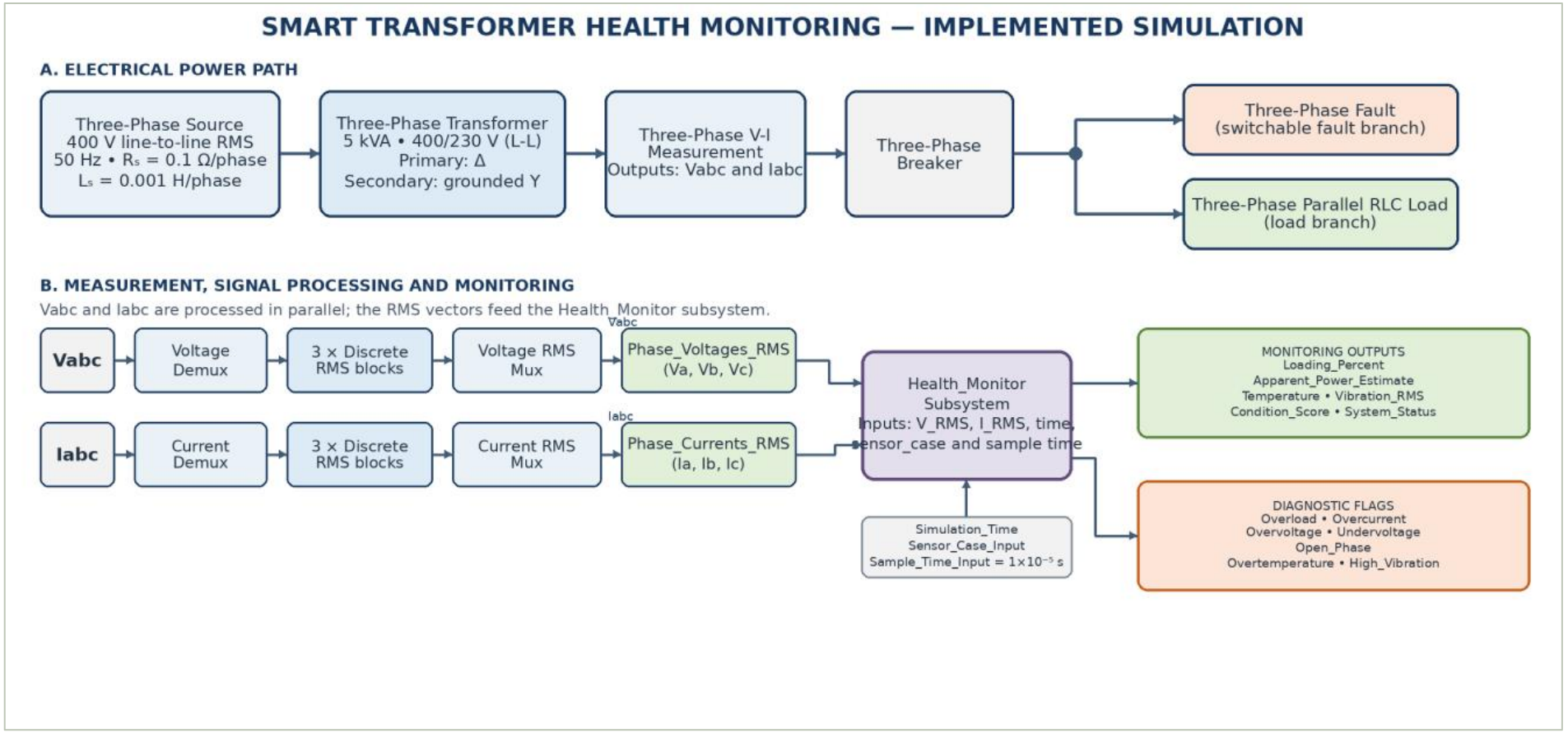
Objectives

1. Model the 5 kVA Δ–Y transformer
2. Monitor phase RMS voltage, current, loading % and apparent power
3. Simulate normal, 150% / 200% overload, over/undervoltage and unbalance
4. Examine open-phase and short-circuit responses
5. Observe temperature, vibration, condition score, status and flags
6. Compare calculated references with displayed results

LITERATURE REVIEW

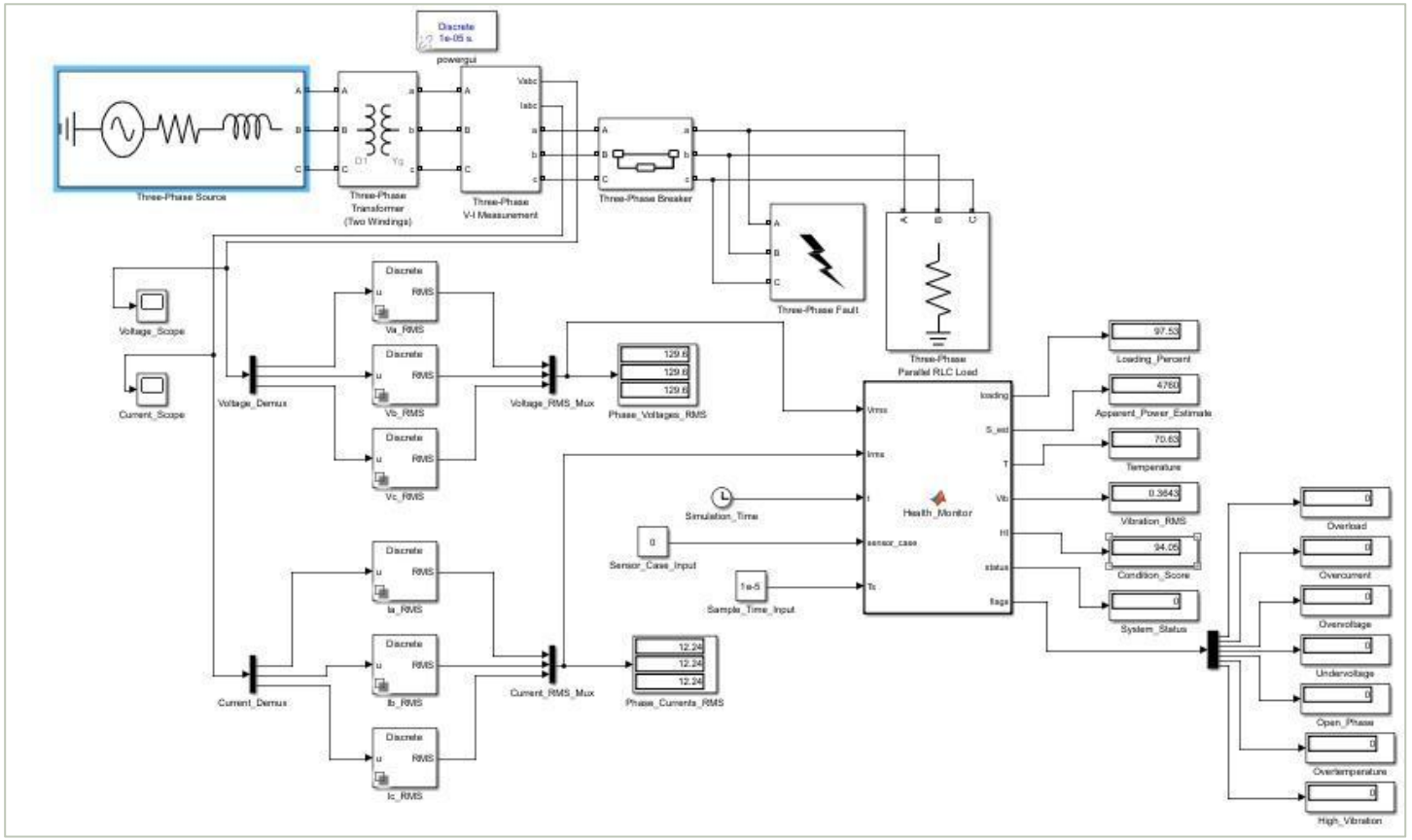
Reference	Main contribution	Application in this project
IEC 60076-1	Defines transformer ratings and requirements	Establishes specifications
MathWorks transformer model	Three-phase transformer simulation	Models the 5 kVA unit
MathWorks sensor documentation	Voltage and current sensor connections	Guides V-I measurement
CIGRE TB 630	Integrated condition monitoring	Guides system architecture
IEEE CALCON 2022 review	Transformer thermal behaviour, hot-spot monitoring	Supports temperature monitoring
Health-index review (2025)	Conventional and ML health assessment	Informs the condition score

PROPOSED SOLUTION: ARCHITECTURE



Three linked sections: three-phase electrical plant, RMS measurement and signal processing, and the Health Monitor subsystem, which outputs condition values and seven diagnostic flags. Figure: functional architecture of the Simulink model.

PROPOSED SOLUTION: SIMULINK MODEL



Complete MATLAB/Simulink transformer monitoring model

12.55 A — Rated secondary line current: 5000 / (√3 × 230)

132.79 V — Secondary phase voltage: 230 / √3

10.58 Ω — Reference load per phase for 5 kW

- Source 400 V L-L, 50 Hz; Rs = 0.1 Ω, Ls = 1 mH
- Stop time 2 s; sample time 1×10^{-5} s
- Loading % = max(Ia, Ib, Ic RMS) / 12.5511 × 100
- S est = 3 · Vph · Iph

PROPOSED SOLUTION: HEALTH MONITOR

Inputs

- Phase RMS voltages
- Phase RMS currents
- Simulation time
- Sensor-case input
- Sample time

Health_Monitor
MATLAB Function subsystem

Indicators

- Loading %
- Apparent power
- Temperature*
- Vibration RMS*
- Condition score
- Status 0 / 1 / 2

7 Flags

- Overload
- Overcurrent
- Over / undervoltage
- Open phase
- Overtemperature
- High vibration

** Model-generated indicators, not physical sensor measurements*

Test conditions simulated

Normal 400 V	**150% overload**	**200% overload**	**440 V over**
360 V under	**Unbalanced load**	**Open phase (0.8 s)**	**Short circuit (0.8–1.0 s)**

PROJECT TIMELINE

1

Model development

Weeks 1–2

Build and run the transformer model

2

Measurement setup

Week 3

Obtain voltage and current readings

3

Simulation and testing

Weeks 4–5

Normal, varying-load and fault cases

4

Result analysis

Week 6

Analyse graphs, record observations

5

Report preparation

Week 7

Compile findings, finish the report

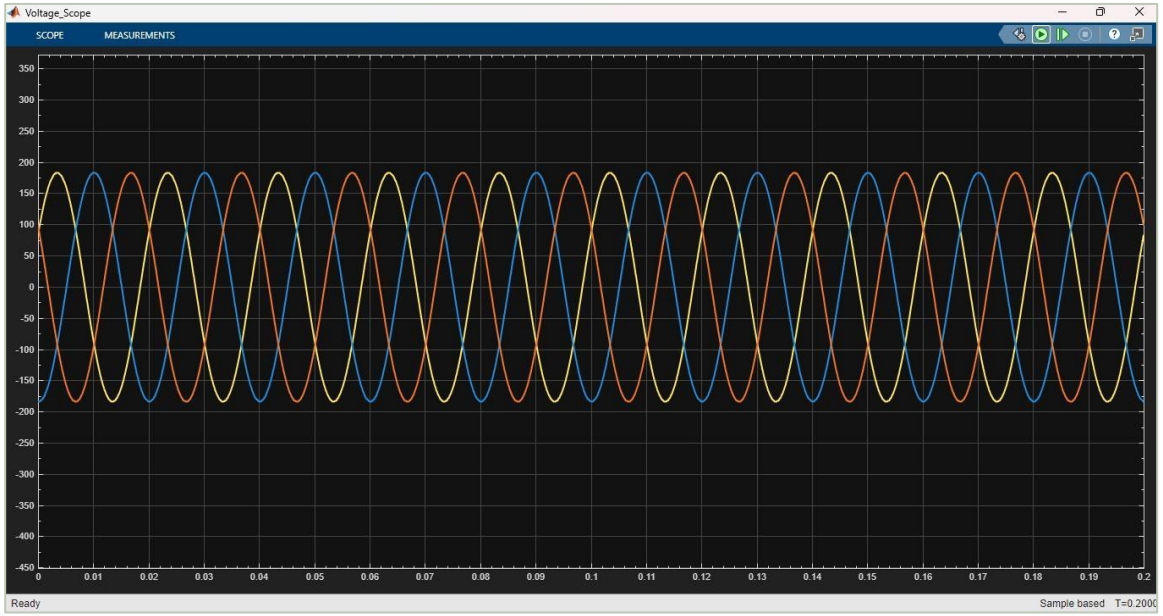
Tools: MATLAB/Simulink. Deliverables: transformer simulation model, voltage and current readings, simulation graphs and observations, final report. Timeline is tentative.

BUDGET

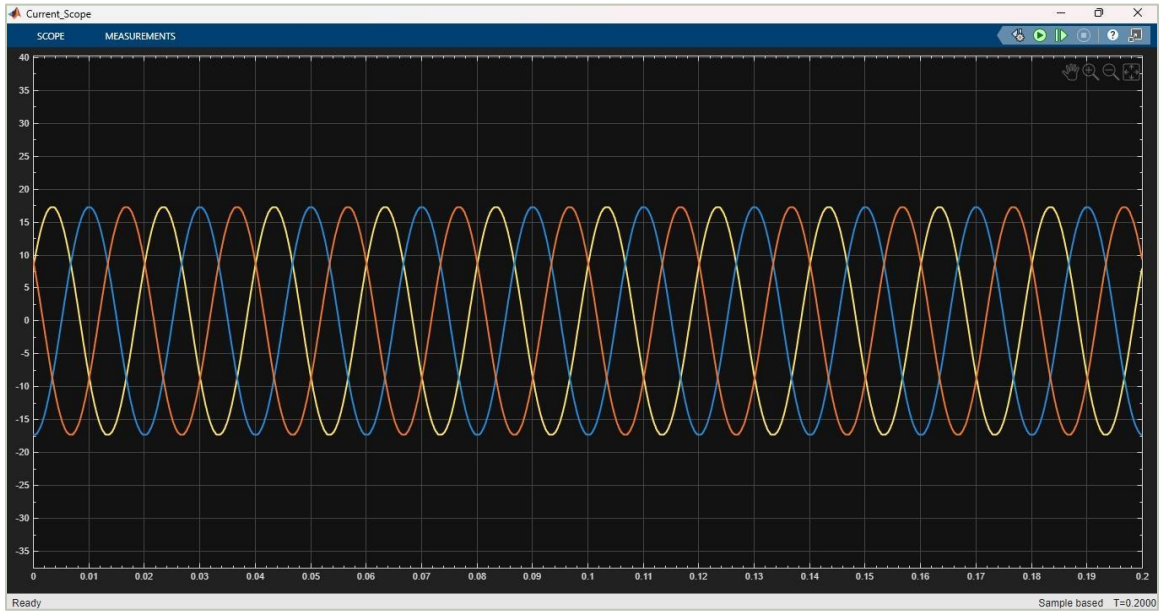
Item	Status this semester
Software: MATLAB, Simulink, Simscape Electrical	Used for all work; licence cost [to be confirmed]
Hardware (sensor board, ESP32)	Not purchased: out of scope for this semester
Documentation and report	No additional cost
Future hardware demonstrator	Cost to be estimated for next semester

The project is simulation-first: no hardware spend is needed before the model is validated.

OUTCOMES: NORMAL OPERATION



Secondary-voltage waveforms

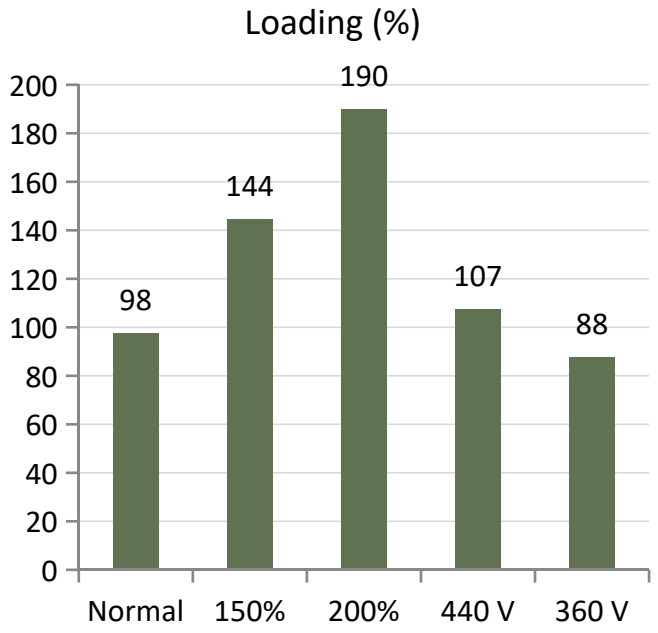


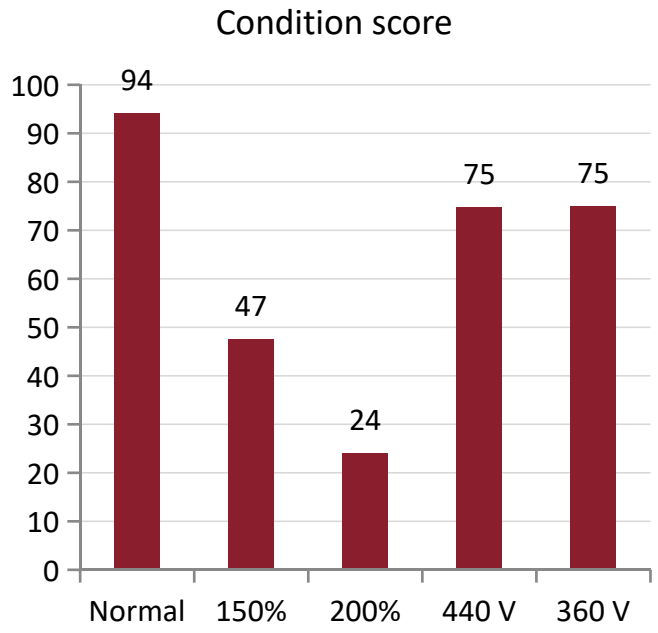
Phase-current waveforms

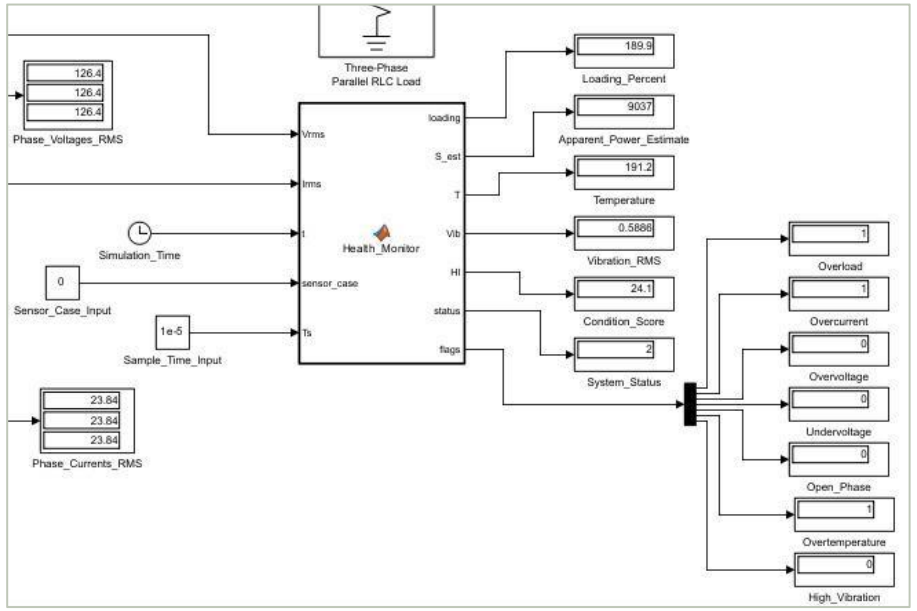
129.6 V	12.24 A	97.53%	4,760 VA	94.05
Phase voltage RMS	Phase current RMS	Loading	Apparent power est.	Condition score

Balanced waveforms, all seven flags 0, status Normal. Voltage is 2.40% and current 2.48% below ideal; cause cannot be assigned without full parameters.

OUTCOMES: LOADING AND VOLTAGE



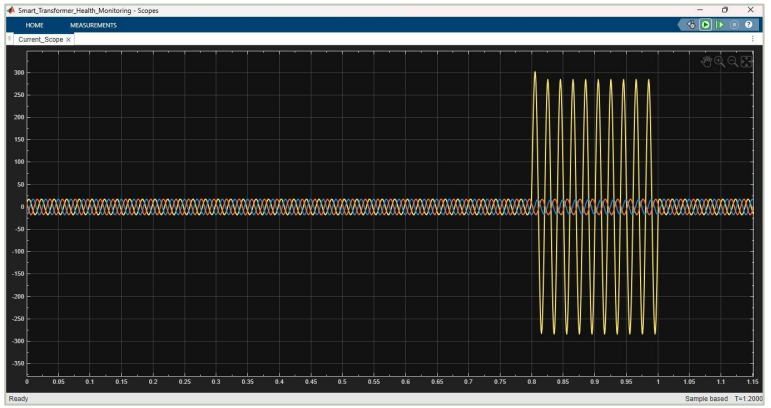




Dashboard at 200% target: overload, overcurrent and overtemperature flags active

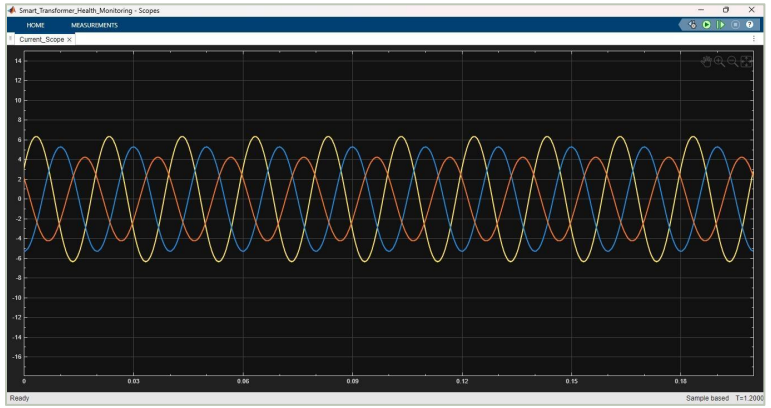
- Lower load resistance (7.05 Ω → 5.29 Ω) raises current 18.12 → 23.84 A and S est 6.96 → 9.04 kVA; overload flag at both levels, overcurrent only at 200%.
- Displayed phase voltage follows the source (116.7 / 129.6 / 142.6 V), about 2.4% below ideal; under- and overvoltage flags raise correctly.

OUTCOMES: FAULT TESTS



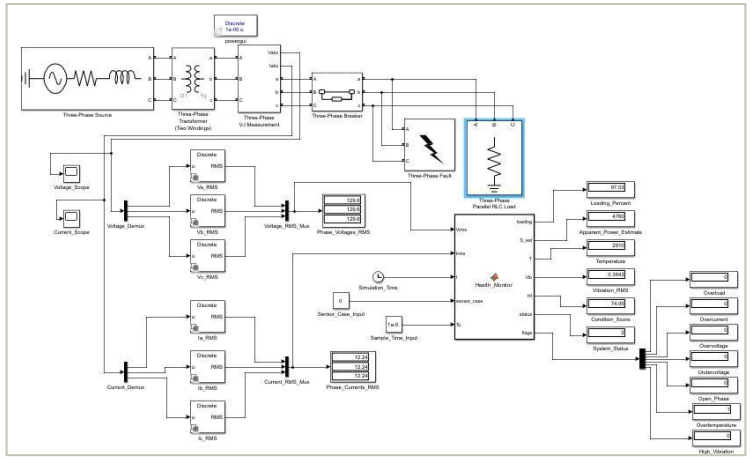
Short circuit

Fault from 0.8 s to 1.0 s gives a large current surge, then returns to normal.



Unbalanced load

Phase currents differ while voltages stay near 132 V; status stays 0.



Temperature proxy

Output reached 2,910 °C in one run: a model artefact, scaling needs rework.

Open phase: Phase A opens after 0.8 s; Ia ≈ 0 A while Ib and Ic stay near 12.2 A. Flag states for open-phase and short-circuit runs are not documented in the report, so none are claimed.

OUTCOMES: CONCLUSION & FUTURE SCOPE

Conclusion

- Model monitors RMS voltage, current, loading and apparent power
- Overload, over/undervoltage and fault cases give identifiable responses
- Calculated and displayed values agree closely
- Temperature and vibration remain proxy indicators

Future scope

- Use real nameplate and test data
- Add sensor noise and quantisation
- Validate thermal models; use accelerometer data
- ESP32 hardware demonstrator and IoT link
- Event logging and validated health scoring

REFERENCES

1. IEC, IEC 60076-1: Power Transformers. webstore.iec.ch/en/publication/588
2. MathWorks, “Two-Winding Transformer (Three-Phase),” Simscape Electrical documentation, R2026b.
3. MathWorks, “Line Voltage Sensor (Three-Phase),” Simscape Electrical documentation, R2026b.
4. CIGRE, Guide on Transformer Intelligent Condition Monitoring (TICM) Systems, TB 630, 2015.
5. “A Comprehensive Review of Recent Developments in Thermal Condition Monitoring of Dry-type Transformers,” IEEE CALCON, 2022. DOI 10.1109/CALCON56258.2022.10060689
6. S. T. Zahra et al., “Power transformer health index and life span assessment,” Eng. Appl. Artif. Intell., vol. 139, 109474, 2025.
7. MathWorks, “Fault (Three-Phase)” and “Power Sensor (Three-Phase),” R2026b.
8. MathWorks, “RMS Measurement,” Simscape Electrical documentation, R2026b.

THANK YOU